DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY

MIDTERM EXAMINATION FALL-2022

Differential Equations (MT-227)

Time Duration :90 Mins **Format-B** **Max Marks: 20**

Question No. 1: CLO 1 [5]

The population of bacteria in a culture grows at a rate proportional to the number of bacteria present at time t. After 3 hours it is observed that 400 bacteria are present. After 10 hours 2000 bacteria are present. What was the initial number of bacteria?

$P_0 = 200$

Question No. 2: CLO 1 [5]

Construct the orthogonal trajectory of the polar curve $r = a(1 + sin\theta)$. $-\ln($

Question No. 3: CLO 2 [5]

Use the appropriate substitution to compute the solution of $(x + ye^{\frac{y}{x}})dx - xe^{\frac{y}{x}}dy = 0,\ \ y(1) = 0.$

Question No. 4: CLO 2 [5]

Compute the solution of $y''' + 3y'' - 4y = 0,\ y(0) = y'(0) = y''(0) = 1$ by applying the method of *undetermined coefficients*

$C_1 =$

$C_2 =$

SEAT NO. CT-030

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR(BACHELOR OF SCIENCE IN
COMPUTER SCIENCE & INFORMATION TECHNOLOGY)
FALL SEMESTER EXAMINATIONS 2021
BATCH 2020

Dated:22-02-2022

Time: 3 Hours

Max.Marks:60

Differential Equations - MT-227

Note: Attempt ALL questions and all questions carry EQUAL marks.

Q1) A) Obtain the solution of the wave equation $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$ using method of separation of variable with all possible cases.

Q1) B) A string is stretched and fastened to two points l apart. Motion is started by displacing the string in the form $y = a\sin\frac{\pi x}{l}$ from which it is released at a time $t = 0$. Find the displacement of any point at a distance x from one end at time t. 6

Q2) A) Suppose that a large mixing tank initially holds $300\ gallons$ of water is which $50\ pounds$ of salt have been dissolved. Another brine solution is pumped into the tank at a rate of $3\ gal/min$, and when the solution is well stirred, it is then pumped out at a same rate. If the concentration of the solution entering is $2\ lb/gal$, determine a differential equation for the amount of salt $A(t)$ in the tank at time $t > 0$.

Q2) B) The number $N(t)$ of supermarkets throughout the country that are using a computerized checkout system is described by the initial-value problem.

$$\frac{dN}{dt} = N(1 - 0.0005N), \quad N(0) = 1$$

How many companies are expected to adopt the new technology when $t = 10$?

Q3) A) Solve the differential equation $x\left(\frac{dy}{dx} + y\right) = 1 - y$ describes the shape of a plane curve C that will reflect all incoming light beams to the same point and could be a model for the mirror of a reflecting telescope, a satellite antenna, or a solar collector.

Q3) B) Find the temperature $T(h)$ in the circular ring is determined by the differential equation 6

$$\frac{d^2T}{dh^2} - 4\frac{dT}{dh} = h^3 e^{2h} - 4T$$

Q4) A) The radial displacement in a rotating disc at a distance r from the axis is given by

$$r^2 \frac{d^2u}{dr^2} + r\frac{du}{dr} + kr^3 - u = 0$$

where k is constant. Solve the equation under the conditions $u = 0$ when $r = 0, u = 0$ when $r = a$.

Q4) B) Find the solution of the partial differential equation which used for discrete-time version of the dynamic programming principle (DPP) for optimization. 6

$$x(z^2 - y^2)\frac{\partial z}{\partial x} + y(x^2 - z^2)\frac{\partial z}{\partial y} = z(y^2 - x^2)$$

Q5) A) Solve the partial differential equation $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6\frac{\partial^2 z}{\partial y^2} = y\cos x$, to determine the motion of a pressure-less fluid subjected to dissipation.

Q5) B) Control system from a simple home heating controller use differential equation to describe the behavior of various closed loop functional blocks. Use the Laplace transform to solve the given initial-value problem:

$$y'' - 4y' = 6e^{3t} - 3e^{-t}, \qquad y(0) = 1, y'(0) = -1$$

Scanned with CamScanner

SEAT NO. CT-18028

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR FALL SEMESTER (SOFTWARE ENGINEERING/
BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY)
EXAMINATIONS 2019
BATCH 2018

Dated:03-02-2020

Time: 3 Hours **Max.Marks:60**

Differential Equations & Linear Algebra - MT-273

- Attempt all questions. All questions carry equal marks

Q.1(a): A 15-volt electromotive force is applied to an LR-series circuit in which the inductance is 0.2 henry and resistance is 40 ohms. Find the current $i(t)$ if $i(0) = 0$. Determine the current as $t \to \infty$. **[CLO-1]** (06)

(b): Find the charge on the capacitor and the current in an LC-series circuit when $E(t) = 100Sin\omega t\ V$, $L = 0.1\ h$, $C = 0.1\ f$, $q(0) = 0\ C$ and $i(0) = 0\ A$. **[CLO-1]** (06)

Q.2(a): Define Cryptography, Plaintext and Ciphertext. Obtain the Hill cipher of the message **I AM SOFTWARE ENGINEER** for the enciphering matrix $\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ **[CLO-1]** (06)

(b) Find the standard matrix for the operator T: $R^3 \longrightarrow R^3$ defined by $T(x,y,z) = (x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta,\ z)$ that first rotates a vector counter clockwise about the Z-axis through an angle 45, then reflects the resulting vector about yz-plane, and then projects that vector orthogonally onto the xy-plane. Draw a sketch of the projects on to the xy-plane and interpret your obtained results **[CLO-1]** (06)

Q.3(a): Solve the given first order ordinary differential equation: **[CLO-2]** (06)

i) $(y^2e^{xy^2} + 4x^3)dx + (2xye^{xy^2} - 3y^2)dy = 0$

ii) $\left[xCos\left(\frac{y}{x}\right) + ySin\left(\frac{y}{x}\right)\right]y - \left[ySin\left(\frac{y}{x}\right) - xCos\left(\frac{y}{x}\right)\right]x\frac{dy}{dx} = 0$

(b): Find the general solution of any one higher order ordinary differential equation: **[CLO-2]** (06)

i) $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = e^{2x}Cos3x + (x^2 + x + 9)$

ii) $x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 5y = x^2Sin(lnx)$

Q.4(a): Form the Partial differential equation of $z = me^{ny}Cosnx$. **[CLO-2]** (02)

(b): Solve the given first order Partial differential equation: **[CLO-2]** (04)

$$y^2\frac{\partial z}{\partial x} - xy\frac{\partial z}{\partial y} - x(z - 2y) = 0$$

(c): Determine whether the vectors a = (1,-2,3), b = (5,6,-1), c = (3,2,1) are linearly independent or linearly dependent in R^3 and interpret your obtained results **[CLO-2]** (06)

Q.5 Find the eigen values and eigen vectors of the matrix A^3 where **[CLO-2]** (12)

$$A = \begin{pmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{pmatrix}$$

----------X----------X----------

Scanned with CamScanner

Dr. Faqiha Sultan

SEAT NO. CT - 005

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR(BACHELOR OF SCIENCE IN COMPUTER SCIENCE & INFORMATION TECHNOLOGY)
FALL SEMESTER EXAMINATIONS 2022
BATCH 2021

Time: 3 Hours

Dated:06-02-2023
Max.Marks:60

Differential Equations - MT-227

Instructions:
Each question carries equal marks.
Solve all the parts together.

Question 1 (CLO 1)

Use appropriate differential models to solve the following problems.

a) A 30-volt electromotive force is applied to an LR series circuit in which the inductance is 0.1 henry and the resistance is 50 ohms. Find the current $i(t)$ if $i(0)=0$, and **estimate** its value as $t \to +\infty$.

b) A thermometer is removed from a room where the temperature is 70° F and is taken outside, where the air temperature is 10° F. After one-half minute the thermometer reads 50° F. With the help of differential model, **predict** the time the thermometer will take to reach 15° F.

Question 2 (CLO 1)

a) **Determine** the orthogonal trajectories of the family of the curves $y = ce^{\frac{-x}{4}}$.

b) Consider the differential equation $y'' - 2\beta y' + (\beta^2 - 1)y = 0$ for the instantaneous position of the vibrating mass with the damping constant β. **Detect** the values of β, if any, for which all solutions tend to zero as $x \to +\infty$.

Question 3 (CLO 2)

a) **Calculate** the solution of the non-homogeneous Cauchy-Euler's differential equation $x^2y'' - 3xy' + 13y = 4 + 3x$.

b) **Formulate** the partial differential equation of all spheres whose center lie on the z-axis and given by the equation $x^2 + y^2 + (z-a)^2 = b^2$.

Question 4 (CLO 2)

a) **Use** Lagrange multipliers to find the general solution of $x(z^2 - y^2)\frac{\partial z}{\partial x} + y(x^2 - z^2)\frac{\partial z}{\partial y} = z(y^2 - x^2)$.

b) **Compute** the complementary function (CF) and Particular Integral (PI) of the linear partial differential equation $\frac{\partial^2 z}{\partial x^2} - 3\frac{\partial^2 z}{\partial x \partial y} + 2\frac{\partial^2 z}{\partial y^2} = e^{x+y} + \cos(x+2y)$.

Question 5 (CLO 2)

a) Obtain Laplace transform of $\cos 4t$ by definition of Laplace transform and **use** the first translation theorem to determine $\mathscr{L}\{(1 - e^{t} + 3e^{-4t})\cos 4t\}$.

b) **Apply** Laplace transform to compute the solution of
$y'' + 4y' + 6y = 1 + e^{-t}, \quad y(0) = 0, \quad y'(0) = 0.$

******

25

Scanned with CamScanner

EE-649

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

FIRST YEAR FALL SEMESTER (ELECTRICAL/ TELECOMMUNICATIONS ENGINEERING)
EXAMINATION 2016
BATCH 2015-16 & PREVIOUS BATCHES

Time: 3 Hours

Dated: 16-11-2016
Max. Marks: 60

DIFFERENTIAL EQUATIONS- MT-227

INSTRUCTIONS: Attempt any **FIVE** questions in all. All questions carry equal marks.

Q.1 (a) Form the ordinary differential equation for $y = Ae^{x} + Be^{2x} + Ce^{3x}$ (04)

(b) Solve any **two** of the following first order ordinary differential equations. (08)

(i) $x.Cos\,x.\dfrac{dy}{dx} + y(x.Sin\,x + Cos\,x) = 1$ (ii) $\dfrac{dy}{dx} = \dfrac{y^3}{e^{2x} + y^2}$

(iii) $(3x^2 + 2xy^2)\,dx + 2y(1+x^2)\,dy = 0$

Q.2 (a) Form the partial differential equation for $z = (a+x^2)(b+y^2)$ (04)

(b) Solve any **two** of the following first order partial differential equations. (08)

(i) $2y(z-3)p + (2x-z)q = y(2x-3)$ (ii) $(y+z)p + (z+x)q = x+y$

(iii) $(x^2 - yz).p + (y^2 - xz).q = z^2 - xy$

Q.3 (a) Solve any **one** of the following higher order differential equations. (06)

(i) $(D^3+1)y = Cos^2\left(\dfrac{x}{2}\right) + e^{-x}$ (ii) $x^3y''' + 3x^2y'' + xy' + y = x + \ln x$

(b) Solve any **one** of the following higher order partial differential equations. (06)

(i) $(D^2 - DD' - 2D'^2)z = (2x^2 + xy - y^2)Sin\,xy - Cos\,xy$

(ii) $(D^2 - 4D'^2)z = \dfrac{4x}{y^2} - \dfrac{y}{x^2}$

Q.4 (a) Solve the following set of simultaneous equations. (06)

$\dfrac{dy}{dx} + y = z + e^{x}$; $\dfrac{dz}{dx} + z = y + e^{x}$

(b) Find the orthogonal trajectories of the family of curves (06)

$3xy = x^3 - a^3$, where 'a' is the parameter.

Q.5 (a) Find $L[Cos\,2t.Sinh\,2t]$ **OR** $L(t^3.e^{-2t}.Cos\,t)$ (04)

(b) Verify initial and final value theorem for $f(t) = 1 + e^{-t}(Sin\,t + Cos\,t)$ (04)

(c) Evaluate by using Laplace transform $\displaystyle\int_0^\infty \frac{(1-Cos\,t)}{t}\,dt$ (04)

Q.6 (a) Find $L^{-1}\left[\dfrac{1}{s^2(s^2+4)}\right]$ **OR** $L^{-1}[Cot^{-1}(s+1)]$ (04)

(b) Solve any **one** of the following differential equations by using Laplace transform method. (08)

(i) $y''' - 3y'' + 3y' - y = x^2.e^{2x}$; $y(0) = 1$, $y'(0) = 0$, $y''(0) = -2$

(ii) $y'' + y = x.Cos\,2x$; $y(0) = 0 = y'(0)$

Q.7 (a) Apply RK method to find an approximate value of y when $x = 1$, given that (04)

$\dfrac{dy}{dx} = x + y^2$ $y(0) = 1$, $h = 0.1$

(b) Find $y(1.9)$, $y(1.8)$ and $y(1.7)$ by using Euler's and modified Euler's method respectively if (08)

$\dfrac{dy}{dx} = x + \sqrt{y}$ $y(2) = 4$, $h = -0.1$

THE END

N.D UNIVERSITY OF ENGINEERING & TECHNOLOGY
FIRST YEAR FALL SEMESTER (ELECTRICAL ENGINEERING)
EXAMINATION 2015
BATCH 2014-15

Time: 3 Hour

Dated: 23-11-2015
Max. Marks: 60

DIFFERENTIAL EQUATIONS - MT-227

Note: Attempt any Five questions out of Seven.

Q1. Solve the following differential equations:

a. $y' = e^{3x+2y}$

b. $xy' - 4y = x^6 e^x$

c. $(cosx)y' + (sinx)y = 1$

d. $2xydx + (x^2 - 1)dy = 0$

Q2. Find the solution to the given Initial-value problem:

a. $(D^2 + D - 12)y = 0, y(2) = 2, y'(2) = 0$

b. $\frac{dy}{dx} = (-2x + y)^2 - 7, y(0) = 0$

Q3. Given $y' = \frac{y-x}{y+x}$ with initial condition $y(0) = 1$; find $y(0.1)$ by Euler Method and Euler's Modified Method. Take h=0.02.

Q4. Find the general solution of the higher order differential equation:

a. $(D^3 + 1)y = \cos(2x - 1)$

b. $(D^4 - 2D^3 + D)y = x^4 + 3x + 1$

c. $(D^3 + D)y = 2x^2 + 4sin4x$

Q5. Solve the following second order differential equations:

a. $x^2y'' - xy' + y = lnx$ C.U

b. $x^2y'' - 3xy' + 5y = x^2 \sin(lnx)$ C.E

c. $x^2y'' - 3xy' + 13y = 5 + 3x$ C.E $x = e^t$ $\ln x = t$

Q6. (1) Form the partial differential equations from:

a. $z = ax + by + a^2 + b^2$

b. $z = f(x^2 - y^2)$

Q6. (2) Solve the partial differential equation:

a. $\frac{\partial^3 z}{\partial x^2 \partial y} + 18xy^2 + \sin(2x - y) = 0$

b. $\frac{\partial^2 z}{\partial x \partial y} = xy$

Q7. Find the Laplace transform of:

a. $sin2t\ sin3t$

b. $e^{2t} + 4t^3 - 2sin3t + 3cos3t$

c. $e^{at} sinbt$

------ X ------